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Business Requirements Document

Date: September 2025

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Project: Banking Loan Insight Dashboard

1. Document Information

Project Name	Banking Loan Insight Dashboard
Document Type	Business Requirements Document (BRD)
Author	Milan Kumar
Version	1.0
Date	September 2025
Tools Used	Power BI, SQL, Microsoft Excel
Status	Completed

2. Executive Summary

Financial institutions managing large loan portfolios face significant challenges in monitoring applicant data, tracking approval rates, and identifying risk patterns across segments. Manual reporting processes are slow, error-prone, and lack the granularity required for strategic decision-making.

This project delivers an end-to-end analytics solution - an interactive Power BI dashboard backed by an SQL-driven ETL pipeline - that consolidates loan application data from multiple sources into a single, trusted source of insight. The solution enables lending managers and analysts to move from reactive reporting to proactive, data-driven portfolio management.

3. Business Problem Statement

The lending operations team lacked a unified, real-time view of their loan portfolio. Key issues included:

- Loan data was spread across multiple disconnected systems with no single source of truth.
- Analysts spent excessive time manually compiling reports instead of interpreting insights.
- No standardized KPI framework existed to measure approval rates, risk levels, or applicant segments.
- Operational teams could not identify approval bottlenecks or high-risk applicant clusters quickly.

4. Business Objectives

- Centralize 600+ loan application records into a validated, analysis-ready dataset.
- Track core lending KPIs across 355,000+ applicants including approval rate, loan status, and risk tier.
- Segment applicants by credit grade, loan purpose, employment length, and region to surface approval trends.
- Reduce operational report generation time by at least 25% through dashboard automation.
- Improve end-to-end Customer Experience (CX) by reducing processing time for loan decisions.

5. Stakeholders

Stakeholder	Role	Interest
Lending Manager	Primary User	Monitor portfolio KPIs and approval rates
Credit Analyst	Primary User	Segment applicants, identify risk patterns
Operations Team	Secondary User	Reduce manual reporting effort
Data Analyst (Milan Kumar)	Developer	Build ETL pipeline and dashboard
IT / Data Team	Support	Data source access and pipeline validation

6. Scope

6.1 In Scope

- ETL pipeline to extract, clean, and validate loan application data using SQL and Excel.
- Power BI dashboard with multi-page KPI tracking: loan summary, applicant segmentation, approval trends.
- SQL-driven applicant segmentation by region, loan grade, and purpose.
- Calculation of a 69.3% overall approval rate and supporting trend metrics.

6.2 Out of Scope

- Real-time data streaming or live database connections (static dataset used).
- Predictive loan approval modelling (handled in a separate ML project).
- Integration with external credit bureau APIs.

7. Business Requirements

Req ID	Requirement Description	Priority	Source
BR-01	System shall consolidate all loan application records into a single validated dataset.	High	Operations Team
BR-02	Dashboard shall display total applications, funded amount, amount received, and average interest rate.	High	Lending Manager
BR-03	Dashboard shall calculate Month-to-Date (MTD) and Month-over-Month (MoM) changes for all KPIs.	High	Lending Manager
BR-04	System shall distinguish Good Loans (Fully Paid, Current) from Bad Loans (Charged-Off) with separate KPI cards.	High	Credit Analyst
BR-05	System shall segment applicants by loan purpose, term, employment length, home ownership, and geography.	Medium	Credit Analyst
BR-06	Dashboard shall provide a filterable details view showing individual loan records.	Medium	Operations Team

BR-07	All data transformations shall be validated against source records before publication.	High	Data Team
BR-08	Dashboard shall reduce manual reporting effort by automating previously manual Excel calculations.	Medium	Operations Team

8. Key Performance Indicators (KPIs)

KPI	Definition	Target
Overall Approval Rate	% of applications with Fully Paid or Current status	Baseline: 69.3%
Total Loan Applications	Count of all applications in the reporting period	355,000+ tracked
Total Funded Amount	Sum of disbursed loan amounts	Tracked monthly
Total Amount Received	Sum of repayments received from borrowers	Tracked monthly
Average Interest Rate	Mean interest rate across all active loans	Tracked by segment
Average DTI	Debt-to-Income ratio across borrower base	Risk monitoring
Good Loan %	Share of portfolio classified as good loans	> 85% target
Processing Time Reduction	Reduction in time to generate operational reports	30% achieved

9. Assumptions & Constraints

Assumptions

- Source data is accurate and complete as extracted from the loan management system.
- Users have access to Power BI Desktop or Power BI Service for dashboard consumption.
- Dataset represents a static snapshot; real-time refresh is not required for this version.

Constraints

- Dataset limited to 600+ validated records due to data access restrictions during the internship period.
- Dashboard must load and filter within 5 seconds on standard hardware.
- No PII (Personally Identifiable Information) to be exposed in any public-facing view.

10. Success Criteria

- All 8 business requirements (BR-01 through BR-08) are implemented and verified.
- Dashboard accurately reflects a 69.3% approval rate consistent with source data.
- Operational processing time reduced by minimum 30% vs. prior manual process.
- Stakeholder review confirms the dashboard supports data-driven lending decisions.